

The Stale Present: Video-Language Models Often Report the Previous State Instead of the Current One

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Abstract

We ask open video-language models (VLMs) a simple question: what is shown at the end of the video? The answer is clearly visible in the final frames, and every model answers it almost perfectly when given only the last frame. Given the whole video, however, many models answer with the previous state of the scene. We call this failure the stale present. On synthetic scenes with four earlier layouts and a 0.5 s final state, accuracy of eight open models ranges from 0.30 to 0.80, against 1.00 from the last frame alone. On real chess games, Qwen3-VL-8B names the captured piece instead of the capturing one in 59–78% of trials. Using controlled stimuli and preregistered tests, we find four things. (i) The effect is driven by the number of earlier states of the same kind and by how briefly the final state is shown; frame rate does not matter. (ii) It is a mid-layer retrieval by the question tokens: blocking the text positions’ attention to the previous state in a narrow band of middle layers removes most stale answers in Qwen3-VL-8B and Gemma-3-12B. (iii) The same holds on real footage once a clean previous state is present; there, real and rendered frames give effects of the same size. (iv) A training-free contrastive decoding that subtracts the score of the video truncated before the last change removes the stale answers. However, it never beats simply reading the last frame on pure present-state questions; it helps only when the question also needs the history. We also report the limits of the phenomenon. The newest model we tested, Qwen3.6-27B, is immune (0.98–1.00). In natural real-world shuffling videos, the effect of history is robust (about 10 points) but only some models concentrate their errors on the previous state. We therefore present the stale present as a documented, mechanistically localized failure of current mid-sized video VLMs, not as a universal property.

1. Introduction

A video assistant is often asked about the *current* state of a scene: whether a drawer is open now, where an object ended

up, which piece now stands on a chess square. Answering needs two things: seeing the last frames and knowing that they are the last ones. Video benchmarks usually conflate these two with memory and reasoning. We isolate them with a deliberately easy question whose answer is plainly visible at the end of the clip, and a matched control that shows the model only the final frame.

The failure we study is easy to state. A model reads the last frame correctly when it sees that frame alone, yet with the full video it names the state that came *just before* (Figure 1). The error is not random: when a model is wrong, it almost always chooses the immediately preceding state. We call this the *stale present*.

This paper is an analysis paper. Most tests were preregistered in a versioned file before the data were generated or the models were run. We report failed predictions and retracted interpretations alongside the supported ones. Our contributions are:

- **A controlled demonstration** (§4) across twelve open VLMs from five families, on synthetic layouts and on real chess games (MET-Bench [5]). Every comparison is paired with a last-frame control, and the real-footage comparisons also with a same-pipeline “frozen video” control.
- **What drives it** (§5). The effect needs at least one earlier state of the same kind as the final one, grows with the number of such states, and shrinks with the final dwell. In a factorial test, going from 1 to 4 prior states adds 0.24 to the last-frame-minus-video gap, longer prior states add 0.07, and frame rate adds nothing.
- **Where it happens** (§6). Blocking the question tokens’ attention to the previous state inside a narrow band of middle layers (relative depth 0.4–0.7) removes most stale answers in two models, and fails in a third. Answer-position “recency heads” found by correlation are not causal.
- **Real footage** (§7). In natural shuffling videos from Perception Test [18], history costs about 10 points of accuracy over a frozen-video control. When a clean, well-held previous state is shown, the stale rate rises by 0.10 on real frames, as much as on rendered tiles, and the same mid-layer masking removes it.

- **A fix and its boundary** (§8). Present-contrastive decoding (PCD) removes the stale answers. It is useful only on questions that need the history as well as the present; for pure present-state questions, reading the last frame is as good.

We are explicit about the limits (§9). The newest model we tested is immune; sample sizes are 50–100 per cell; and the strongest real-footage effect uses a spliced timeline.

2. Related work

State tracking in video VLMs. VET-Bench [15] shows that VLMs are at chance on the shell game and that grounded trajectory reasoning fixes it. MET-Bench [5] finds a large gap between text-based and image-based entity tracking in chess and shell games. TOC-Bench [4] and StateTrace [8] test object identity and hidden-state persistence under occlusion. Event bookkeeping at high frequencies fails as well [3]. Our question is simpler than all of these: the answer is visible, nothing is occluded, and no tracking is needed.

Last-frame shortcuts and temporal grounding. Several works note that single frames often match or beat full clips on video QA [16]. EgoIntent [17] reports that the last boundary frame outperforms the ordered clip for several models. VLX-VR [13] observes models being drawn to a later salient event instead of the state a question asks for. We study the reverse direction, an earlier state overriding the current one. We control for image quality with a frozen-video condition, and we localize the effect inside the model.

Contrastive decoding. Contrastive decoding [14] and its visual variants [10] subtract the scores of a degraded input. SEASON [20] builds temporally homogenized negatives for video. Our PCD uses the video truncated before the last detected change as the negative, which targets exactly the stale state. We find that its value is narrower than such methods usually claim.

Localizing retrieval by attention blocking. We follow the attention-knockout methodology of Geva et al. [7]: we block attention edges from text positions to chosen visual tokens in chosen layers and measure the change in the answer.

3. Setup

Models. We evaluate the following models locally in bf16:

- Qwen3-VL 2B/4B/8B/32B-Instruct [1], Qwen3.5-9B, Qwen3.6-27B and Qwen2.5-VL-7B [2], all with native video input;
- LLaVA-OneVision-7B [11] and InternVL3.5-8B [19], with video input (at most 32 uniformly sampled frames, always including the last);

- Gemma-3-12B [6], Idefics3-8B [9] and InternVL3.5-GPT-OSS-20B-A4B, with frames given as a numbered image list (“Frame i :”), at most 32 uniformly sampled frames, always including the last.

LLaVA-NeXT-Video-7B was excluded because it could not read the final frame (last-frame accuracy 0.22–0.74). Qwen models receive every frame; their video encoders merge two consecutive frames into one temporal token group. Local inference used four A100-40GB GPUs.

Questions and scoring. Every question is multiple choice with three options: the correct final content, the content of the previous state at the queried place (the *stale* option), and one other content present in the video. For the three-color layouts the options are simply the three colors in a fixed order; elsewhere the option order is shuffled. Exact templates are in Appendix A. We score the argmax over the option-letter logits right after the generation prompt, with thinking disabled. We report accuracy and P(stale), the rate of choosing the stale option.

Controls.

- *Last frame*: the final frame alone, as a single image, with the same question.
- *Frozen video* (real footage only): the final frame repeated for the length of the clip and fed through the same video pipeline. It keeps the video path’s resolution and temporal merging but removes the history.

The difference between the frozen video and the real video isolates the effect of history.

Stimuli.

- **Synthetic layouts.** Colored cups or disks on a table, rendered in 2D at 4 fps. Static layouts change abruptly; there is no motion and no occlusion. A reveal layout is followed by m further layouts, each held 1 s, and the last one is held D_{fin} .
- **Real chess.** Board positions from real MET-Bench game records, rendered with glyphs, $m \in \{1, 3\}$ prior moves, $D_{\text{fin}} \in \{0.5, 2\}$ s. On capture squares, the stale answer is the captured piece.
- **Word tiles.** Letter tiles spelling the orders of the real Perception Test word-shuffle videos, used for the factorial study.
- **Real footage.** Perception Test letter-shuffling videos (via MVBench [12]) with the official per-second letter boxes (§7).

Figure 2 shows one example of each. In all synthetic and chess stimuli the changes are abrupt: there is no motion, no hand and no occlusion, so the final state is fully visible from its first frame. Construction details are in Appendix B.

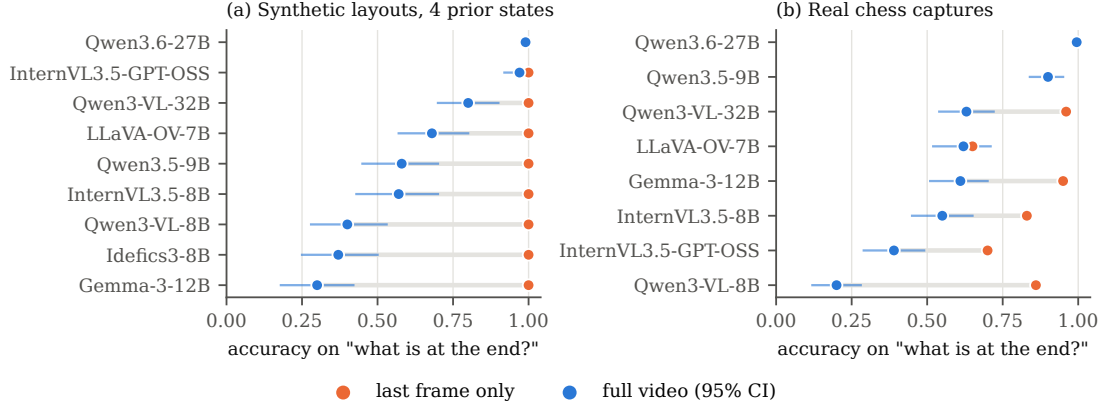
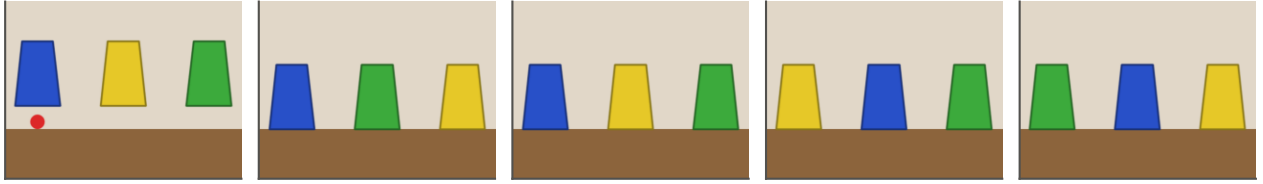
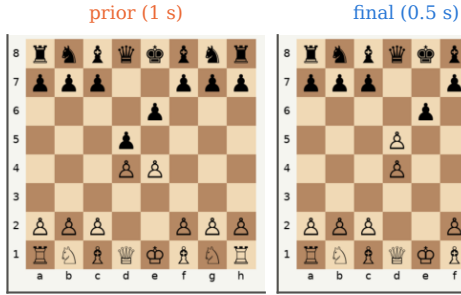


Figure 1. **The stale present across models.** Accuracy on “what is at [position] at the very end?” when the model sees the last frame only (orange) or the full video (blue, 95% bootstrap CI). (a) Synthetic colored layouts with four prior layouts and a 0.5 s final layout ($n = 60$ per model). (b) Real chess games, capture squares, one prior move, 0.5 s final position ($n = 100$). For Qwen3.6-27B only the video condition was run. In (b), Idefics3-8B is omitted because it cannot read the board (last-frame accuracy 0.48).

(a) Synthetic layouts, $m = 4$. Q: “At the very end, what color is the cup in the right position?” Correct: yellow; stale: green.
 reveal (2.5 s) prior 1 (1 s) prior 2 (1 s) prior 3 (1 s) final (0.5 s)



(b) Real chess game, capture move.
 Q: “At the very end, what is on square d5?”
 Correct: white pawn; stale: black pawn.



(c) Real Perception Test clip and its rendered-tile twin.
 Q: “At the end, which letter is in the first place?”
 Correct: C; stale: A.

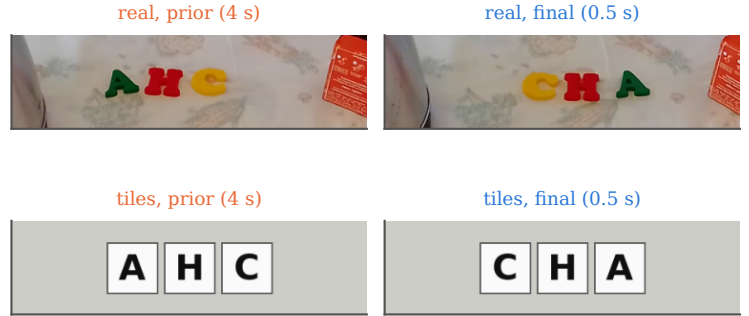


Figure 2. **Example stimuli.** One frame per state; durations in parentheses. The previous state is in orange and the final state in blue. (a) Synthetic cups: a reveal layout (the ball is visible) is followed by $m = 4$ layouts with abrupt changes. (b) A real game from MET-Bench, rendered as boards; the stale answer is the captured piece. (c) A Perception Test clip reduced to a clean previous resting state (4 s) and the final state (0.5 s), cropped to the letters, and the same timeline rendered as tiles (§7). With the full video, Qwen3-VL-8B answers “A” on this clip; from the last frame, or the final frame repeated as a video, it answers “C”.

Statistics. We use paired bootstrap 95% CIs (10,000 resamples; stratified by model when pooling) and exact McNemar tests. The CI table for all fix experiments is Holm-corrected within each difference column. Each preregistration states a threshold, and we call a prediction supported only if the

point estimate meets it. Two real-footage tests (§7) had CIs clearly above zero but point estimates of 0.096 and 0.097 against a threshold of 0.10; we report them as failed.

4. The phenomenon

Main result. Figure 1 shows the core comparison. On synthetic layouts with $m = 4$ and $D_{\text{fin}} = 0.5$ s, every model answers the last frame alone perfectly, yet video accuracy is 0.30 (Gemma-3-12B) to 0.80 (Qwen3-VL-32B). Qwen3-VL-2B and 4B score 0.42 and 0.45. The errors are stale: for Qwen3-VL-8B and Qwen3.5-9B with $m = 2$, 52–59 of 55–70 errors pick the immediately preceding layout. On real chess captures, Qwen3-VL-8B reads the final board at 0.86 but answers from the video at 0.20, choosing the captured piece in 78% of trials. The paired last-frame minus video differences are highly significant for Qwen3-VL-8B on every chess cell (McNemar $p \leq 10^{-7}$ after Holm correction).

Not every model, and not the newest. Qwen3.6-27B scores 0.98–1.00 on every synthetic and chess cell. InternVL3.5-GPT-OSS is almost immune on synthetic layouts (0.97) but not on chess (0.39, with P(stale) of 0.57). Qwen3.5-9B shows no gap on the easiest chess cell. Within the Qwen3-VL family, the 32B model is much less affected than the 2B–8B models.

5. What drives the stale present

Final dwell. In the first test, three layouts followed the reveal, with the dwells of the last two varied independently (Qwen3-VL-8B and Qwen3.5-9B, $n = 60$ per cell). Accuracy rises from 0.45–0.58 at $D_{\text{fin}} = 0.5$ s to 0.83–0.93 at 4 s. The dwell of the previous layout has only a small effect.

Same-kind prior states. Table 1 varies the number m of layouts after the reveal. When the only earlier layout is the visually distinct reveal (cups lifted, ball visible; $m = 1$), the present is reported correctly even at 0.5 s (0.92–1.00). One additional cups-down layout ($m = 2$) halves accuracy (0.48–0.57), and the errors are almost all one step back: 52–59 of the 55–70 errors pick the immediately preceding layout. Going from $m = 2$ to $m = 4$ changes little on these layouts. Showing a single initial layout for 2.5–5 s before a 0.75 s final layout does not produce the effect either (0.87–0.98). The model does not simply weight states by how long they were shown. It confuses *which of several similar states is the last one*.

Minimal displays. A single digit or a single ball that jumps between positions does not trigger the effect (0.92–1.00). Three colored disks do, when asked “where is the red disk?” (Qwen3-VL-8B 0.50, Qwen3.5-9B 0.48). A single disk also triggers it when the question is keyed by position (“what is on the left?”): Qwen3-VL-8B 0.54, Qwen3-VL-32B 0.46. The stale present thus needs either several objects or a position-keyed query.

Table 1. **Prior layouts, final dwell and input format** (synthetic cups, $n = 60$ per cell). Accuracy from the video, with P(stale) in parentheses; “images”: the same frames given as a timestamped image list. The last frame alone gives 1.00 in every cell.

m	D_{fin}	Qwen3-VL-8B		Qwen3.5-9B	
		video	images	video	images
1	0.5 s	0.92 (.02)	–	1.00 (.00)	–
2	0.5 s	0.57 (.43)	0.85	0.48 (.47)	0.75
2	1 s	0.70 (.28)	–	0.68 (.28)	–
2	2 s	0.82 (.18)	0.92	0.77 (.23)	0.82
4	0.5 s	0.40 (.43)	0.70	0.58 (.35)	0.82
4	2 s	0.73 (.18)	0.93	0.75 (.17)	0.83

Not a tokenization artifact. Qwen models merge two frames into one temporal token group. All state changes in our videos fall on group boundaries, so no group mixes old and new frames. The per-frame spatial grid is the same (20×28 patches) in video and image mode. Feeding the same frames as a timestamped image list weakens the effect (Table 1; Qwen3-VL-8B, $m = 2$: 0.57→0.85) but does not remove it; P(stale) at $D_{\text{fin}} = 0.5$ s stays at 0.12–0.23. Models that take image lists natively (Gemma-3, Idefics3) show the effect in full (Figure 1a).

Factorial test of the timeline. On letter-tile renderings of the real word-shuffle items, we crossed three factors for five models ($n = 67$ each): the number of prior states $m \in \{1, 4\}$, the dwell per prior state $\in \{1, 4\}$ s, and the frame rate $\in \{2, 4\}$ fps. The final state was held 0.5 s. Figure 3 shows the result. The main effect of m on the last-frame minus video gap is +0.240 [0.181, 0.300] for position questions and +0.236 [0.187, 0.284] for order questions. The dwell adds +0.069 [0.046, 0.094]; frame rate adds −0.011 [−0.027, 0.004]. P(stale) follows the same pattern. At $m = 4$, every model shows a position-question gap of 0.42–0.60, including Qwen3.5-9B, whose gap at $m = 1$ is only about 0.04 (per-model values in Table A4). Two of the three preregistered directions were wrong: we expected higher frame rates and *shorter* prior states to hurt, and neither did. On the tiles, going from one to four same-kind prior states roughly doubles the pooled gap. On the cup layouts, one and three same-kind prior states ($m = 2$ and $m = 4$ in Table 1) give similar accuracy. We do not know what causes this difference.

Other interventions. Appending the final frame as an extra image after the video, with the text “The last frame of the video is shown above”, only partly helps (Qwen3-VL-8B, $m = 4$: 0.40→0.65; Qwen3.5-9B: 0.58→0.87; the last frame alone gives 1.00). We had preregistered that this anchoring would restore at least 0.9 in every cell; it did not.

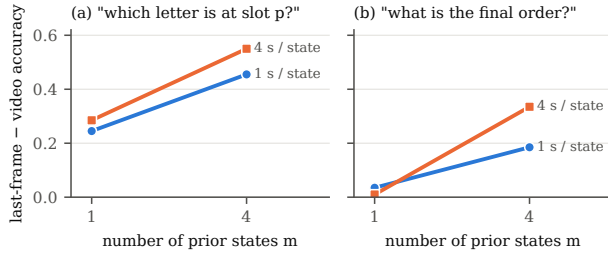


Figure 3. **More prior states, and longer ones, make the present staler.** Pooled last-frame minus video accuracy (5 models, $n = 67$ each; averaged over 2 and 4 fps, which had no effect). The final state is held 0.5 s. (a) Position questions. (b) Order questions, which need all letters: with a single prior state they show no gap.

The video overrides an explicitly given present. Chain-of-thought is inconsistent: it helps some cells, hurts others, and the written reasoning often describes intermediate layouts that never appeared.

Summary of the drivers. The stale present is not a general weakness in reading the end of a video: single objects that jump are reported correctly. It arises when (i) several states of the same kind occur, (ii) the question selects a place or an object whose content differs between the last two of them, and (iii) the final state is brief relative to the history. Under these conditions the model knows the candidate contents but not which of them is current.

6. Where the stale answer is retrieved

Attention is not recency-weighted, and correlated heads are not causal. In Qwen3-VL-8B, the final and the previous layouts receive similar attention from the answer position (final share 0.46–0.53). A few heads in layers 24–27 attend to the final layout more on correct trials (0.76–0.79) than on stale ones (about 0.50). We preregistered a causal test: suppress the previous layout for these heads at the answer position. It had no effect, so the prediction was falsified.

Band masking. We instead block *all* text positions after the video from attending to the previous state’s visual tokens, only within a chosen band of layers. The mask is applied inside the attention function; visual positions are untouched. Figure 4a shows the drop in $P(\text{stale})$ per band.

- **Qwen3-VL-8B** (36 layers; dwell data, $D_{\text{fin}} \leq 1$ s). Masking in all layers raises accuracy from 0.60 to 0.80 and lowers $P(\text{stale})$ from 0.33 to 0.075. Layers 18–26 alone give the full effect (0.81, 0.08). Layers 0–8 and 27–35 do nothing. Masking only the last token barely helps (0.60→0.63), so the retrieval is spread over the question tokens.

- **Gemma-3-12B** (48 layers; preregistered, $n = 320$). Masking in all layers raises accuracy from 0.547 to 0.709 (McNemar 54:2, $p = 4 \times 10^{-14}$). The single band 18–23 gives 81% of this gain, and the 8 global-attention layers alone give about 95%. Masking the final state instead collapses accuracy to 0.138 ($P(\text{stale})$ 0.728). Masking earlier states changes accuracy by only -0.03 .
- **Idetics3-8B.** All-layer masking gives only 0.45→0.50, so the preregistered prediction fails.
- **Qwen2.5-VL-7B.** Partial: the best band is 14–20 of 28, but $P(\text{stale})$ drops only from 0.37 to 0.28.

In the two models where masking works, the stale answer is fetched by the question tokens from the previous state’s visual tokens in middle layers. The remaining gap to the last-frame control (0.81 vs 1.00 in Qwen3-VL-8B) presumably sits in the visual token stream itself. For Idetics3 and Qwen2.5-VL, more of the effect seems to sit there.

7. Real footage

Natural videos: history costs accuracy, but errors are not reliably stale. We used the Perception Test videos in which letters are shuffled and the question asks for the final word order. A preregistered pilot (6 models, 79 questions) failed: the pooled last-frame minus video difference was $+0.04$ [$-0.00, 0.08$], against a threshold of 0.10. PCD also failed, because its change detector fires on hand and camera motion. We then diagnosed the failure step by step:

- **Final dwell is not the cause.** Cropping to the letters and cutting the clip 0.5 s or 2 s after the last change did not help ($+0.01$).
- **Real shuffles are multi-step.** Reconstructed from the per-second letter boxes, 42 of 74 videos contain 3–13 changes, and the initial order is the true previous state in only 10 of 73. The pilot’s “stale option” was therefore mostly the wrong target.
- **History has a robust cost.** With the question targeting the reconstructed previous state, and with a frozen-video control that goes through the same pipeline, history costs $+0.097$ [$0.051, 0.144$] accuracy (position questions, 6 models). The pipeline itself costs only $+0.025$ (n.s.).
- **But the errors are not reliably stale.** $P(\text{stale})$ rises from the frozen video to the real video only for InternVL3.5-8B (0.18→0.31) and LLaVA-OV-7B (0.21→0.31).

A further check explained this. Most reconstructed “previous states” are transit orders captured while a hand carries a letter. Counting only *resting* states (no letter centroid moves more than $0.25 \times$ the letter height between adjacent samples), 45 of 73 videos have exactly one resting prior state, and 22 have none. Natural shuffles therefore offer essentially $m = 1$, the regime in which even synthetic videos show a small effect (Figure 3).

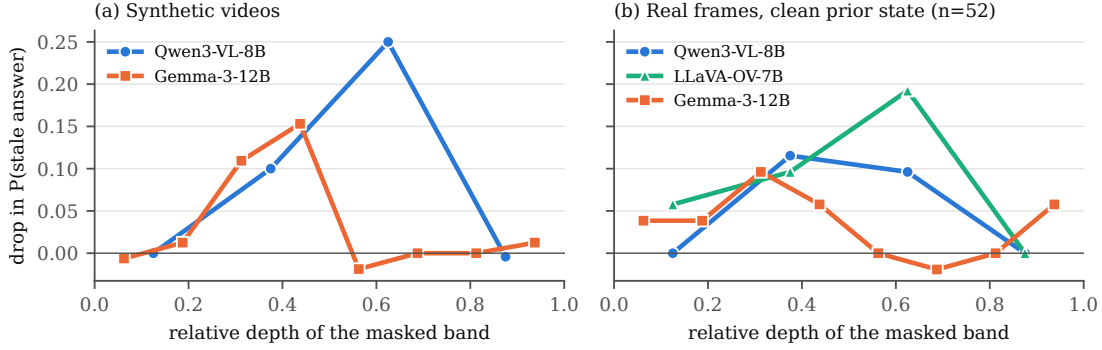


Figure 4. **The stale answer is retrieved in middle layers.** Drop in $P(\text{stale})$ when the question tokens are blocked from the previous state’s visual tokens within one band of layers (x: band center divided by the number of layers). (a) Synthetic videos: Qwen3-VL-8B ($n = 240$, $D_{\text{fin}} \leq 1$ s) and Gemma-3-12B ($n = 320$). (b) Real frames with a clean prior state held 4 s (§7; $n = 52$ per model). Shallow and deep bands have no effect.

Real frames on a controlled timeline. For the 52 videos with a resting prior state, we built clips at 2 fps: a real frame of that state held 4 s, then the real final frame for 0.5 s. The same timeline was also rendered with letter tiles. Table 2 shows the result. History raises $P(\text{stale})$ by $+0.103$ [0.054, 0.151] on real frames and by $+0.106$ on tiles, so the pre-registered prediction on $P(\text{stale})$ is supported. The accuracy prediction failed narrowly ($+0.096$ against the threshold of 0.10). The domain difference is $+0.035$ [−0.022, 0.093]. Real images do not block the stale present; natural timelines rarely set it up. The largest real-frame effects are in Qwen3-VL-8B (0.21→0.40) and LLaVA-OV-7B (0.17→0.42). Order questions show no effect in either domain, as expected at $m = 1$.

The same mid-layer retrieval on real frames. We applied band masking to these real clips (preregistered; Figure 4b, Table 3). In all three models, the unmasked answers reproduced the earlier run exactly.

- **Qwen3-VL-8B.** Blocking the previous state in all layers lowers $P(\text{stale})$ from 0.40 to 0.25 (9:1, $p = 0.02$). This is close to its no-history level of 0.21.
- **LLaVA-OV-7B.** $P(\text{stale})$ falls from 0.42 to 0.21 (12:1, $p = 0.003$), and the single band 14–20 of 28 gives almost all of it.
- **Gemma-3-12B.** The real-frame effect is small; only its global layers give a significant drop.

We had predicted that Qwen3-VL-8B’s synthetic band (18–26) would also be the best band on real frames. It recovered 62% of the effect, but the adjacent band 9–17 did slightly better, by one item. The localization thus holds at the level of “middle layers” but is broader and slightly shallower on real frames.

8. A training-free fix and where it stops working

Method. We first locate the last change t_c with a pixel change detector. *Present-contrastive decoding* (PCD) then scores each option as

$$s(a) = \log p(a \mid \text{video}) - \alpha \log p(a \mid \text{video}_{<t_c}),$$

with $\alpha = 1$ fixed in the preregistration. The second term measures what the model would answer from the past alone, so the stale option is penalized. For Qwen3-VL we also test *change-point masking* (CPM), which applies the mid-layer mask of §6 to all visual tokens before t_c .

Pure present-state questions: no better than the last frame. PCD removes most stale answers (Table 4). For example, Qwen3-VL-8B on chess goes from 0.20 to 0.84, with $P(\text{stale})$ from 0.78 to 0.04, and CPM performs similarly. But reading the last frame alone is just as good. Across 88 model×cell comparisons (9 models), PCD is never significantly above the last frame after Holm correction. In 8 of them it is significantly below, e.g. Gemma-3 on layouts (−0.27) and Idefics3 on layouts (−0.25). If a question asks only about the present, the right fix is to answer from the last frame.

Questions that need the history. PCD earns its keep when the answer must combine history and present. We asked, on real chess exchanges: “What is *now* on the square where the first move landed?” The question does not name the stale answer. The last frame alone cannot identify the square, and the video alone answers stale. PCD improves over the last frame by $+0.29$ to $+0.47$ for Qwen3-VL-8B/32B, Qwen3.5-9B and InternVL3.5-8B (Holm-corrected $p \leq 0.007$ in both dwell cells). At a 0.5 s final dwell it also helps Gemma-3 ($+0.35$) and InternVL3.5-GPT-OSS ($+0.36$). It does not

Table 2. **Real frames vs. rendered tiles on the same timeline** (prior state 4 s, final 0.5 s; 6 models $\times$ 52 items, position questions). Frozen: the final frame repeated through the video pipeline.

	last frame	frozen	video	frozen — video	P(stale): frozen $\rightarrow$ video
Real frames	0.72	0.70	0.60	+0.096 [0.048, 0.144]	0.22 $\rightarrow$ 0.33
Rendered tiles	0.92	0.94	0.80	+0.131 [0.096, 0.170]	0.04 $\rightarrow$ 0.15

Table 3. **Band masking on real frames** ($n = 52$ per model). P(stale) (accuracy in parentheses). Counts: items that stopped / started being stale; McNemar p .

mask	Qwen3-VL-8B	LLaVA-OV-7B	Gemma-3-12B
none	0.40 (0.58)	0.42 (0.58)	0.35 (0.56)
previous state, all layers	0.25 (0.69); 9:1, $p=.02$	0.21 (0.73); 12:1, $p=.003$	0.25 (0.63); 6:1, $p=.13$
best single band	0.29 (0.65), L9–17	0.23 (0.69), L14–20	0.23 (0.63), global layers
final state, all layers	0.83 (0.10)	0.75 (0.17)	0.52 (0.27)
frozen video (no history)	0.21	0.17	0.29

help LLaVA-OV or Idefics3. A disk version of the same question fails for every model and method (at most about 0.6): the models cannot retrieve the initial position either.

Robustness and failures.

- **The change point must be estimated.** Truncating a fixed Δ seconds hurts whenever Δ is shorter than the final dwell (InternVL3.5, 2 s dwell: 0.63 $\rightarrow$ 0.35).
- **Noise.** Under simulated camera noise, the pixel detector finds only 19 of 520 change points. A jitter-compensated detector (brightness normalization and ± 3 px alignment; threshold set on a disjoint development set) restores the gains.
- **Real footage.** On natural video, hand and camera motion make the detector place the last change near the end of the clip (median 98% of its length). PCD then fails.
- **Questions that name the stale item.** PCD is harmful when the question itself names the stale item (one cell fell from 0.82 to 0.05).
- **Noise amplification.** It can amplify noise in a model that ignores the input: Idefics3’s single-ball control fell from 0.44 to 0.04.

9. Limitations and discussion

The newest model is immune. Qwen3.6-27B shows no stale present on any of our tests. Qwen3-VL-32B is much less affected than the 8B model, and InternVL3.5-GPT-OSS is almost immune on synthetic layouts. We did not test closed frontier models. The stale present may therefore be a property of current mid-sized open models that newer training already removes, rather than a lasting limitation. We think the controlled measurements, the timeline factors and the localized retrieval are still useful. They give a concrete test for new models, a diagnosis for the models that fail, and

a warning for benchmark design: a clip whose final state is short and preceded by similar states can penalize a model for retrieval, not for perception.

Which models escape, and why? We can only describe the pattern. Within Qwen3-VL, the effect shrinks from 2B–8B to 32B, and the later Qwen3.6-27B shows none. InternVL3.5 with a GPT-OSS language model is almost immune on layouts but chooses the captured piece in 43–57% of chess trials. Feeding frames as a timestamped image list helps the Qwen models without removing the effect, and models that only take image lists (Gemma-3, Idefics3) are strongly affected. Neither scale, nor the vision encoder, nor the input format alone explains who escapes. We did not have access to training data, so we cannot test whether the newer models saw more present-state or temporal-order supervision.

What the controls buy. Two controls changed our conclusions. The last-frame control showed that most proposed fixes for pure present-state questions, including our own PCD, do not beat simply answering from the last frame. The frozen-video control showed that part of the apparent real-footage gap is caused by the video pipeline itself (lower per-frame resolution, temporal merging), not by history: for Qwen3-VL-8B in the natural videos, nearly all of the gap was a pipeline effect. We recommend that any claim of the form “watching the video hurts the answer about its end” be paired with both controls.

Implications for video QA benchmarks. Clips whose last state is short and preceded by similar states test more than perception: a model can see the answer and still retrieve the previous state. Benchmarks that ask about the end of a

Table 4. **Training-free fixes.** Accuracy: last frame / video / PCD ($\alpha = 1$). Chess and layouts ($m = 4$) with $D_{\text{fin}} = 0.5$ s. History questions: “what is now on the square where the first move landed?”, $D_{\text{fin}} = 2$ s, $n = 100$. The last-frame value there is unpaired.

model	chess captures, $m = 1$	layouts, $m = 4$	history question (chess)
Qwen3-VL-8B	0.86 / 0.20 / 0.84	1.00 / 0.40 / 0.87	0.38 / 0.51 / 0.82
Qwen3-VL-32B	0.96 / 0.63 / 0.98	1.00 / 0.80 / 0.98	0.46 / 0.50 / 0.88
Qwen3.5-9B	0.90 / 0.90 / 0.98	1.00 / 0.58 / 0.88	0.41 / 0.70 / 0.88
InternVL3.5-8B	0.83 / 0.55 / 0.76	1.00 / 0.57 / 0.87	0.39 / 0.63 / 0.78
Gemma-3-12B	0.95 / 0.61 / 0.93	1.00 / 0.30 / 0.73	0.45 / 0.45 / 0.64

clip can separate the two failures by reporting last-frame and frozen-video scores alongside the video score. Conversely, questions that need both the history and the present, such as our history-referenced chess questions, cannot be solved from the last frame and are where methods like PCD add value.

Scale and realism. Most cells have $n = 50$ – 100 , and the real-frame experiments have 52 items per model. Band-level comparisons within a model differ by only a few items and should not be read as layer-exact. Our synthetic scenes are clean 2D renders, and the chess boards are rendered from real game records. The strongest real-footage effect comes from a spliced timeline. In natural videos we find a robust cost of history but not a reliable preference for the previous state.

Preregistration. Hypotheses and thresholds were written into a versioned file before each experiment. We did not use an external registry, so the ordering is documented only by our own records. Several analyses are explicitly exploratory: the non-Qwen replications of the fixes, the band splits in the natural-video diagnosis, and the global-layer finding in Gemma-3.

10. Conclusion

Asked what a video shows at its end, many current open video VLMs answer with what it showed just before. The failure grows with the number of similar earlier states and with a brief final state, and it does not depend on frame rate. It is produced by a mid-layer retrieval from the question tokens to the previous state’s visual tokens, and it can be removed at inference time. It appears on real frames as strongly as on rendered ones once a clean previous state is present. The newest model we tested does not show it. For pure present-state questions, the last frame is the strongest baseline, and any fix should be compared against it.

Reproducibility. Code, stimulus generators and the pre-registration file are available at <https://github.com/jsjtsty/StateRev-VL>.

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Appendix

A. Question templates and scoring

All questions end with “Answer with only the letter.” Options are written inline as “(A) ... (B) ... (C) ...”. The video (or image list) comes before the text in a single user turn.

- **Cup layouts:** “The video shows three cups of different colors (blue, green, yellow). At the very end of the video, what color is the cup in the {left, middle, right} position? (A) blue (B) green (C) yellow.”
- **Disks,** position-keyed: “The video shows colored disks in three positions (left, middle, right) that are rearranged several times. At the very end of the video, what is in the {position} position?” Object-keyed: “... where is the {color} disk?” with options Left/Middle/Right.
- **Single digit / single ball:** “The video shows a sequence of digits. What digit is shown at the very end of the video?” and “The video shows a red ball that jumps between positions. Where is the ball at the very end of the video?”
- **Chess:** “The video shows a chess game, one position after another. At the very end of the video, what is on square {square}?” Options are piece names such as “white pawn”.
- **History-referenced chess:** “... At the very end of the video, what is on the square where the first move of the video ended?”
- **History-referenced disks:** “... At the very end of the video, what color is the disk in the position where the {color} disk was at the beginning of the video?”
- **Perception Test:** the original MVBench order question (“What is the order of the letters at the end?”), or our position question “At the end, which letter is in the {first, second, ...} place of the letter sequence?”

Controls. For the last-frame control, “The video shows” becomes “The image shows” and references to the end of the video are removed (e.g. “The image shows a chess board. What is on square {square}?”). For Perception Test the prefix is “The image shows the last frame of a video.” For image-list input, each frame is preceded by “Frame i :” (Gemma-3, Idifics3, InternVL3.5-GPT-OSS) or “Frame i (t s):” (the Qwen image-mode test).

Scoring. We run one forward pass and read the logits at the first generated position. The score of an option is the maximum logit over the tokenizer’s variants of its letter (e.g. “A” and “A”). Thinking is disabled for Qwen3.5 through the chat template. No sampling is used anywhere except in the chain-of-thought test.

B. Stimulus construction

Cups and disks. A 2D side-view renderer draws three cups (or disks) on a table at 448×336 pixels and 4 fps. The reveal layout lifts the cups and shows the ball. Each later layout is a random permutation of the colors that differs from the previous one; the queried position is chosen so that its content differs between the last two layouts. Non-final layouts are held 1 s. Seeds are fixed per dataset.

Chess. We take move windows from the MET-Bench-Chess evaluation games (real MillionBase games) and render each position as a 448×448 board with piece glyphs and file/rank labels, at 4 fps. A window shows m moves; non-final positions are held 1 s. For the capture questions, the queried square is the destination of the last move and that move is a capture, so the previous content is the captured piece. We first also queried source squares (correct answer “empty”), but the models almost never answer “empty” even from the final board (0.04–0.10), so those cells were dropped as an option bias. For the history-referenced questions we select windows of exactly two moves in which the second move recaptures on the square where the first move landed.

Word tiles. For each Perception Test letter item, the letters are drawn as black-on-white tiles on a gray background, in the order of the corresponding state. For the factorial study, prior states are orders of the same letters that differ from their neighbors; the last prior state differs from the final order at the queried place.

C. Real-footage pipeline

Source. We use the Perception Test videos that ship with MVBench and the official Perception Test annotations (`all_valid.json`), which provide per-second bounding boxes for each letter and action segments. We keep the letter-order questions (79 in the pilot).

Crop and settle frame. The crop is the union of all letter boxes over the video, padded by 50% of its size on each side, cut from the 1920×1080 source and resized to 448 px width. The settle frame is the last end frame of any action segment whose objects include a letter.

Resting states. From the per-second boxes we compute each letter’s centroid. A sample is *at rest* if no letter centroid moves more than $0.25 \times$ the median letter height from the previous sample and to the next one. Consecutive resting samples with the same left-to-right order form a run; runs with a repeated order are merged. An item is kept if the last run spells the correct final word (reading direction chosen by the larger spread of the final centroids). Of 73 reconstructable items, 45 have exactly one resting prior state, 7 have two and 22 have none. Without the resting criterion, 42 of 74 videos contain 3–13 order changes, most of them transit orders while a hand carries a letter.

Controlled clips. For the 52 items with a resting prior state, the prior frame is the crop frame closest to the middle of the prior run, and the final frame is the crop frame at the end of the video. The held clip repeats the prior frame 8 times and appends the final frame (2 fps: 4 s then 0.5 s). The frozen clip repeats the final frame 9 times. The tile clips follow the same timeline.

D. Band masking

Implementation. We register a custom attention function (a wrapper around PyTorch SDPA) with the Transformers attention interface. In the chosen language-model layers, it sets the attention mask to $-\infty$ from every query position after the last visual token to every key position of the chosen state’s visual tokens. Vision-encoder layers and visual query positions are never modified. For Qwen3-VL on synthetic videos we used eager attention with the same mask.

Token spans. For Qwen models, the video grid gives T temporal groups of equal size; in the real clips groups $0..T-2$ are the prior state and group $T-1$ the final state. For image-list models (Gemma-3, Idefics3), each frame’s image tokens are located by the image token id. For LLaVA-OneVision, the video tokens are split evenly over the 9 frames, excluding one trailing newline token. In the synthetic datasets, state boundaries come from a pixel change detector (a frame starts a new state if more than 0.1% of its pixels change by more than 40 gray levels).

Bands. Layers are split into 4 equal bands (36- and 28-layer models) or 8 bands (Gemma-3-12B, 48 layers). Gemma-3 has one global-attention layer every six layers (8 in total) and sliding-window layers otherwise; we also mask only the global or only the sliding layers. Unmasked answers reproduced the earlier unmasked runs exactly for all three models on real frames, which checks that the wrapper does not change the computation when no mask is set.

Gemma-3-12B on synthetic stimuli. Table A1 gives the full band results behind Figure 4a. Idefics3-8B: no mask 0.45, previous state masked in all layers 0.50 (29:13, $p = 0.02$); the preregistered gain of 0.15 is not reached. Qwen2.5-VL-7B: no mask 0.39 (P(stale) 0.37); layers 14–20 0.44 (0.28); all layers 0.45 (0.25).

Table A1. **Band masking in Gemma-3-12B** (layouts $m = 4$ and chess captures, $n = 320$).

mask	accuracy	P(stale)
none	0.547	0.378
previous state, all layers	0.709	0.194
final state, all layers	0.138	0.728
earlier states, all layers ($n = 220$ paired)	0.518 $\rightarrow$ 0.486	–
previous state, layers 12–17	0.644	0.269
previous state, layers 18–23	0.678	0.225
previous state, any other 6-layer band	0.528–0.559	unchanged
previous state, 8 global layers only	0.700	0.209
previous state, 40 sliding-window layers only	0.603	0.319

E. PCD details

Change detectors. The pixel detector marks the last frame whose mean absolute difference to the previous frame exceeds 2% of the maximum intensity. The motion-compensated detector (used under noise) normalizes each frame’s brightness, aligns it to the previous frame by the integer shift within ± 3 px that minimizes the difference, blurs, crops an 8 px border, and flags a change when more than 0.4% of the pixels differ by more than 25 gray levels. The threshold was chosen on a development set disjoint from the test sets (every 20th non-capture chess item and the $m = 2$ layouts); the detector matches the clean change point on 36 of 40 development videos. The detector we had preregistered (blurred difference against a median/MAD threshold) found no changes on the noisy development videos and was replaced before any test run.

Noise model. Gaussian pixel noise ($\sigma = 6$), a per-frame random translation of ± 2 px and a per-frame brightness flicker of $\pm 3\%$, seeded per item.

Detector-free truncation. Removing a fixed last $\Delta \in \{0.5, 1\}$ s instead of detecting the change works when Δ is at least the final dwell and hurts otherwise, because the “past” then contains part of the present.

A gate. We also tested a gate that applies PCD only when the truncated video already predicts the full-video answer (“the answer looks like an echo of the past”). For the four models whose runs completed, it kept 98–100% of PCD’s gain on the stale cells. It did not prevent Idefics3’s failure on the single-ball control. Because two preregistered models did not finish, we report the gate only descriptively.

Table A2. **PCD for image-list models** (exploratory). Last frame $\rightarrow$ video (P(stale)) $\rightarrow$ PCD $\alpha = 1$. Idefics3 cannot read the chess board.

cell	Gemma-3-12B	Idefics3-8B	InternVL3.5-GPT-OSS
chess captures, $m = 1, 0.5$ s	0.95 $\rightarrow$ 0.61 (.37) $\rightarrow$ 0.93	–	0.70 $\rightarrow$ 0.39 (.57) $\rightarrow$ 0.79
chess captures, $m = 3, 2$ s	0.98 $\rightarrow$ 0.70 (.29) $\rightarrow$ 0.91	–	0.85 $\rightarrow$ 0.56 (.43) $\rightarrow$ 0.78
layouts, $m = 4, 0.5$ s	1.00 $\rightarrow$ 0.30 (.48) $\rightarrow$ 0.73	1.00 $\rightarrow$ 0.37 (.43) $\rightarrow$ 0.75	1.00 $\rightarrow$ 0.97 $\rightarrow$ 0.98
disks, object-keyed, 0.5 s	1.00 $\rightarrow$ 0.58 (.34) $\rightarrow$ 0.82	0.86 $\rightarrow$ 0.36 (.30) $\rightarrow$ 0.92	1.00 $\rightarrow$ 0.90 $\rightarrow$ 0.98
single disk, position-keyed, 0.5 s	0.72 $\rightarrow$ 0.46 $\rightarrow$ 0.58	0.76 $\rightarrow$ 0.52 $\rightarrow$ 0.54	1.00 $\rightarrow$ 0.48 (.52) $\rightarrow$ 1.00
control: single digit	0.94 $\rightarrow$ 1.00 (PCD)	0.72 $\rightarrow$ 1.00 (PCD)	0.94 $\rightarrow$ 0.96 (PCD)
control: single ball	0.50 $\rightarrow$ 0.66 (PCD)	0.44 $\rightarrow$ 0.04 (PCD)	1.00 $\rightarrow$ 1.00 (PCD)

Table A3. **PCD under camera noise** (native $\rightarrow$ PCD with the motion-compensated detector, $\alpha = 1$). With the plain pixel detector, PCD is at or below native in every cell for Qwen3-VL-8B.

cell	Qwen3-VL-8B	InternVL3.5-8B
chess captures, $m = 1, 0.5$ s	0.39 $\rightarrow$ 0.76	0.53 $\rightarrow$ 0.71
layouts, $m = 4, 0.5$ s	0.38 $\rightarrow$ 0.92	0.53 $\rightarrow$ 0.93
history-referenced chess, 2 s	0.46 $\rightarrow$ 0.61	0.62 $\rightarrow$ 0.72

F. Per-model results

G. Preregistered predictions and outcomes

Table A6 lists every prediction behind the results in this paper, in the order in which they were registered. “Held” means the point estimate met the registered threshold; near misses are marked as failed. Earlier entries in the same file (A–K) concern a different question, tracking hidden objects in the shell game, and are not part of this paper.

Table A4. **Timeline factorial, per model** (word tiles, position questions, last frame minus video accuracy, $n = 67$ per cell; final state 0.5 s).

fps / prior dwell	$m = 1$				$m = 4$			
	2 / 1 s	2 / 4 s	4 / 1 s	4 / 4 s	2 / 1 s	2 / 4 s	4 / 1 s	4 / 4 s
Qwen3-VL-8B	+0.37	+0.30	+0.31	+0.22	+0.48	+0.48	+0.43	+0.57
Qwen3.5-9B	+0.03	+0.06	+0.03	+0.04	+0.45	+0.54	+0.42	+0.57
InternVL3.5-8B	+0.31	+0.30	+0.22	+0.27	+0.43	+0.60	+0.33	+0.60
LLaVA-OV-7B	+0.45	+0.46	+0.40	+0.49	+0.55	+0.57	+0.52	+0.57
Gemma-3-12B	+0.12	+0.33	+0.18	+0.36	+0.43	+0.52	+0.49	+0.52
pooled	+0.26	+0.29	+0.23	+0.28	+0.47	+0.54	+0.44	+0.56

Table A5. **Real footage, per model** (position questions). Left: natural clips cut 0.5 s after the last change, question targeting the reconstructed previous state ($n \approx 72$). Right: controlled clips with a resting prior state held 4 s ($n = 52$). Accuracy for last frame / frozen video / video; P(stale) frozen $\rightarrow$ video.

model	natural clips		controlled clips, real frames	
	accuracy	P(stale)	accuracy	P(stale)
Qwen3-VL-8B	0.67 / 0.62 / 0.61	0.25 $\rightarrow$ 0.25	0.79 / 0.77 / 0.58	0.21 $\rightarrow$ 0.40
Qwen3-VL-32B	0.67 / 0.71 / 0.61	0.21 $\rightarrow$ 0.26	0.71 / 0.69 / 0.65	0.21 $\rightarrow$ 0.25
Qwen3.5-9B	0.74 / 0.68 / 0.60	0.21 $\rightarrow$ 0.19	0.77 / 0.69 / 0.62	0.23 $\rightarrow$ 0.29
InternVL3.5-8B	0.68 / 0.68 / 0.50	0.18 $\rightarrow$ 0.31	0.63 / 0.69 / 0.63	0.23 $\rightarrow$ 0.25
LLaVA-OV-7B	0.72 / 0.67 / 0.54	0.21 $\rightarrow$ 0.31	0.83 / 0.77 / 0.58	0.17 $\rightarrow$ 0.42
Gemma-3-12B	0.61 / 0.57 / 0.49	0.31 $\rightarrow$ 0.25	0.62 / 0.58 / 0.56	0.29 $\rightarrow$ 0.35

Table A6. **Preregistered predictions.**

id	prediction	outcome
L1	Accuracy increases with the final dwell.	Held (0.45–0.58 at 0.5 s, 0.83–0.93 at 4 s).
L1'	P(stale) > 0.1 even with a 4 s final dwell.	Failed (0.07–0.08).
M1	$m = 4$ below $m = 1$ by > 0.15 at 0.5 s.	Held, but $m = 4 \approx m = 2$: a step, not a graded effect.
M2	Image-list input shows the same trends, P(stale) > 0.2 .	Partly: the effect is reduced, P(stale) 0.12–0.23.
N1	Single digit / single ball show P(stale) > 0.2 .	Failed (0.92–1.00 accuracy).
O1	Three disks worse than one by > 0.15 .	Held for object-keyed questions; position-keyed questions fail even with one disk.
P1	Appending the last frame restores ≥ 0.9 .	Failed (e.g. 0.40 $\rightarrow$ 0.65).
Q1, Q3	Correlated “recency heads” at the answer position are causal.	Failed: no effect of either intervention.
R1	Chess video below last frame by > 0.15 , P(stale) > 0.2 .	Borderline on destination squares; a capture-only follow-up (exploratory) shows 0.20 vs 0.86.
R2	$m = 1$ closer to the last frame than $m = 3$.	Failed.
S1	Change-point masking: chess +0.2 and P(stale) < 0.15 .	Partly: accuracy +0.3 to +0.5, P(stale) 0.25–0.28.
S2	PCD +0.1 averaged over sets.	Held for the three Qwen models.
S3	No harm on single-item controls.	Held.
T1, T2	Questions naming the captured piece: stale answers, fixed by PCD.	Failed; the design was confounded (naming the piece cues that it left), and PCD was harmful.
U1	History-referenced questions: P(stale) > 0.25 in ≥ 3 of 5 models.	Held (0.27–0.47 in all five).
U2	PCD beats video and last frame by > 0.15 .	Held for 4 of 5 models on chess; failed on disks.
V1	Fixed-window truncation keeps $\geq 70\%$ of the gain.	Partly: only when the window covers the final dwell.
W1	Pixel-detector PCD collapses under noise.	Held.
W2	A robust detector keeps $\geq 70\%$ of the gain.	Held (after the detector amendment, before test runs).
X1–X3	A gate keeps PCD’s gain and removes its harm.	Incomplete (4 of 6 models); reported descriptively.
Y1	Gemma-3 / Idefics3: masking the previous state +0.15.	Held for Gemma-3 (+0.16); failed for Idefics3 (+0.05).
Y2	One band of 1/8 depth gives most of the gain.	Held for Gemma-3 (layers 18–23, 81%).

id	prediction	outcome
Y3	Masking the final state collapses accuracy.	Held (0.138).
Z1	Natural real footage: last frame minus video ≥ 0.10 .	Failed (+0.04).
AA1, AA2	The same with a short final dwell; dwell dependence.	Failed (+0.01).
AB1	Real frames on a static timeline: gap ≥ 0.10 .	Failed (+0.01).
AC1	Tiles, order questions: gap ≥ 0.10 .	Failed (+0.03).
AC2	Tiles, position questions: gap ≥ 0.10 .	Held (+0.24).
AD	$m = 4 > m = 1$; 4 fps > 2 fps; 1 s prior > 4 s prior.	m held (+0.24); fps failed (≈ 0); prior dwell failed in the opposite direction (4 s prior states widen the gap by 0.07).
AE1	Question at the reconstructed previous state: gap ≥ 0.10 .	Held (+0.12).
AE2	Errors concentrate on the previous state.	Failed (-0.02).
AF1	Frozen minus video ≥ 0.10 .	Failed narrowly (+0.097, CI [0.051, 0.144]).
AF2	Errors concentrate on the previous state vs frozen.	Failed (-0.07).
AG1	Controlled real clips: frozen minus video ≥ 0.10 .	Failed narrowly (+0.096, CI [0.048, 0.144]).
AG2	P(stale) rises with history.	Held (+0.103, CI [0.054, 0.151]).
AH1	Real clips: masking the previous state lowers P(stale) by ≥ 0.10 .	Held (0.40 $\rightarrow$ 0.25, $p = 0.02$).
AH2	Layers 18–26 recover $\geq 50\%$ and are the best band.	Failed: 62%, but layers 9–17 are better by one item.
AH3	Masking the final state raises P(stale).	Held in all three models.